Classical solution for solving the Deutsch Problem

$f : \{0,1\} \to \{0,1\}$

$\Rightarrow$ balanced or constant

Problem: Determine $\boxed{f(0) \oplus f(1)}$

2 queries

$f(x)$

1st query ($x=0$, $f(0)$) — 1st query ($x=0$)

$f(0) = 0$ | $f(0) = 1$

$f(0)=0$
$f(1)=0$

2nd query — 2nd query

$f(0) = 0$, $\boxed{f(1) = 0}$	$f(0) = 0$, $\boxed{f(1) = 1}$	$f(0) = 1$, $f(1) = 0$	$f(0) = 1$, $f(1) = 1$
constant	balanced	balanced	constant

```
if f(0) == 0:
    if f(1) == 0:
        return "Constant"
    if f(1) == 1:
        return "Balanced"
if f(0) == 1:
    if f(1) == 0:
        return "Balanced"
    if f(1) == 1:
        return "Constant"
```

Motivation (quantum)

Can we solve the Deutsch Problem in one query?

classical solution $\Rightarrow$ simple, trivial
$\Rightarrow$ 2 queries $f(0) \oplus f(1)$

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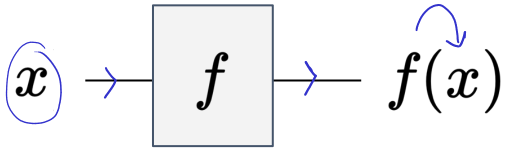
Functions as oracles

$U_f \to |0\rangle, |1\rangle \to \alpha|0\rangle + \beta|1\rangle, \ |\alpha|^2 + |\beta|^2 = 1$

$$f(x) : \{0,1\} \to \{0,1\}$$

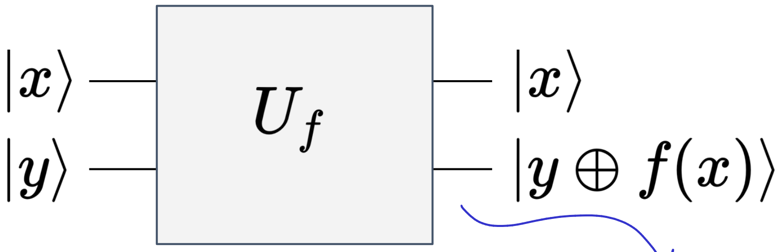


$f(x) = 0$
$f(x) = 1$
$f(x) = x$
$f(x) = \bar{x}$

- Model functions in the quantum circuit model
  - a function, in general, is not reversible
  - quantum operation is unitary and reversible
- By linearity, once we define the operation on basis states, we can extend it to any input state $|\psi\rangle \in (\mathbb{C}^2)^{\otimes 2}$

i/p o/p f(x)

$$|x, y\rangle \to |x, y \oplus f(x)\rangle$$



oracle gate

unitary and **reversible** $U_f \in \mathcal{U}\left((\mathbb{C}^2)^{\otimes 2}\right)$

2x2 I X

$$U_f = \begin{pmatrix} 1-f(0) & f(0) & 0 & 0 \\ f(0) & 1-f(0) & 0 & 0 \\ 0 & 0 & 1-f(1) & f(1) \\ 0 & 0 & f(1) & 1-f(1) \end{pmatrix}$$

I X

$|x, y\rangle \to |x, y \oplus f(x)\rangle$

- Verify that U_f is a Hermitian matrix
- Verify that U_f is a unitary matrix
- Determine U_f for all the cases of $f(x)$

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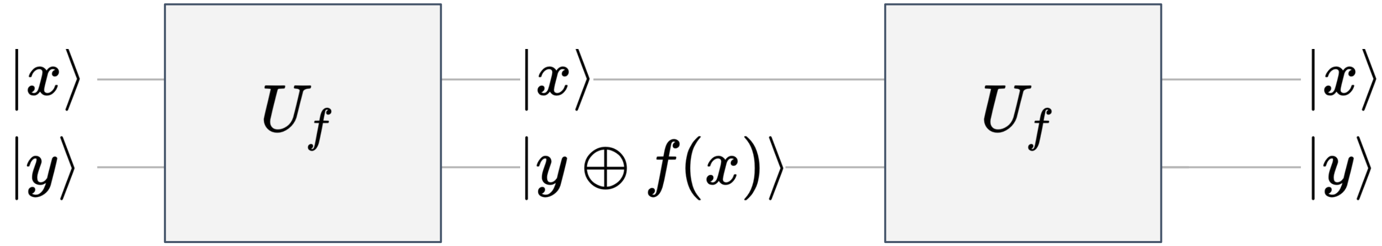
Functions as oracles

$|x,y\rangle \xrightarrow{U_f} \cdot \xrightarrow{U_f} |x,y\rangle$

unitary, reversible.

$|x,y\rangle \xrightarrow{U_f} |x, y \oplus f(x)\rangle \xrightarrow{U_f} |x, (y \oplus f(x)) \oplus f(x)\rangle$



$$|x, (y \oplus f(x)) \oplus f(x)\rangle = |x, y \oplus (\underbrace{f(x) \oplus f(x)})\rangle = |x, \underbrace{y \oplus 0}_{y}\rangle = |x, y\rangle$$

- It is trivial to prove that U_f is self-inverse. Hence this quantum circuit is reversible.
- The oracle gate U_f can also be considered as function-controlled-NOT gate.

$|x,y\rangle \longrightarrow |x, y \oplus f(x)\rangle$

$f(x)=0$

$|x,y\rangle$

$f(x)=1$

$|x,\bar{y}\rangle$

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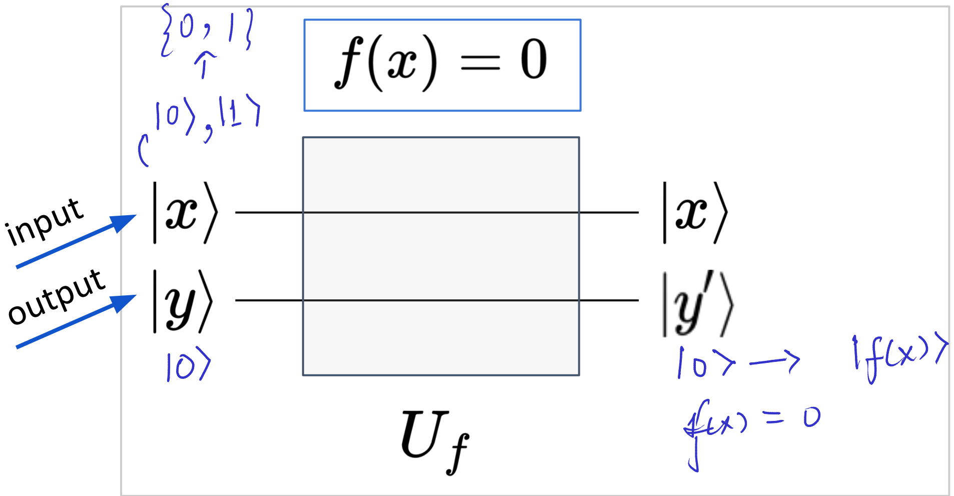
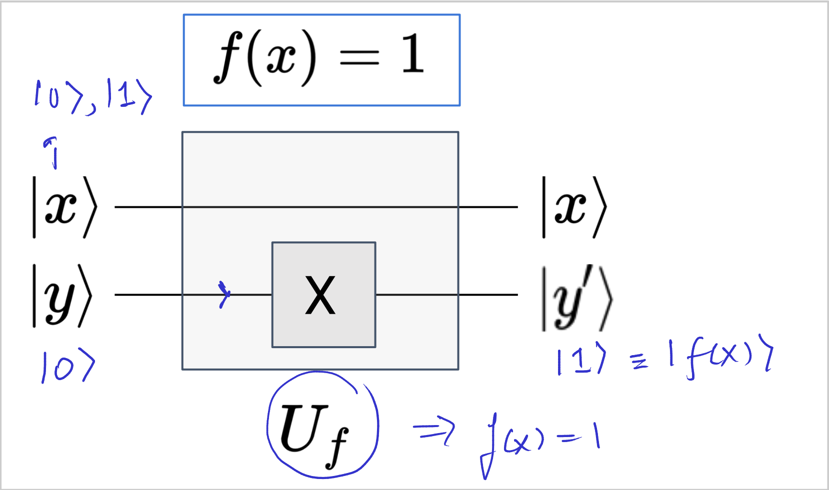
Quantum circuit U_f to implement $f(x) \rightarrow U_f$

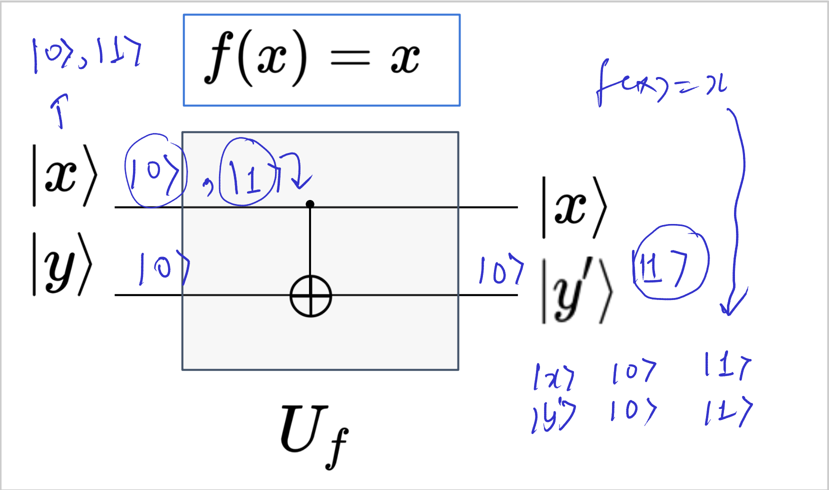
$f: \{0,1\} \rightarrow \{0,1\}$

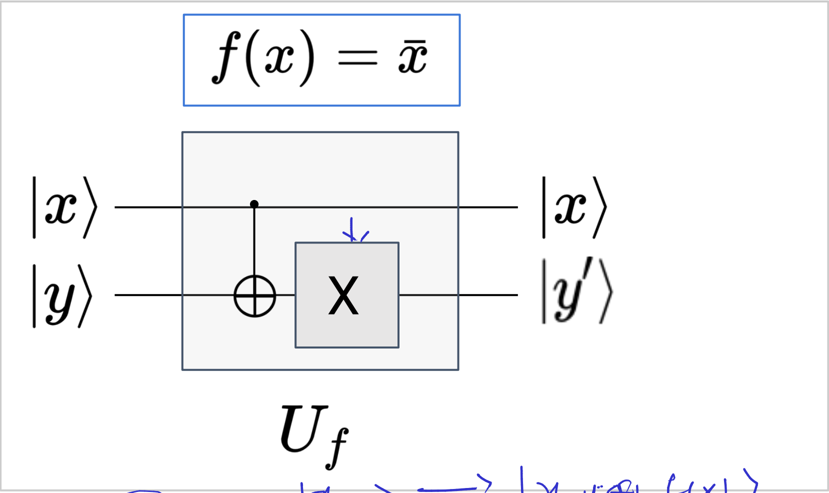
$|x, y\rangle \rightarrow |x, y \oplus f(x)\rangle$

$|x, 0\rangle \rightarrow |x, f(x)\rangle$









$f(x) = \bar{x}$

$\lvert x\rangle$	$\lvert y'\rangle$
$\lvert 0\rangle$	$\lvert 1\rangle$
$\lvert 1\rangle$	$\lvert 0\rangle$

$\rightarrow |x, y\rangle \longrightarrow |x, y \oplus f(x)\rangle$

The universality of quantum computing implies that any function $f(x)$ can be efficiently encoded in this way, to desired accuracy, as a quantum circuit consisting of gates from a finite universal set.

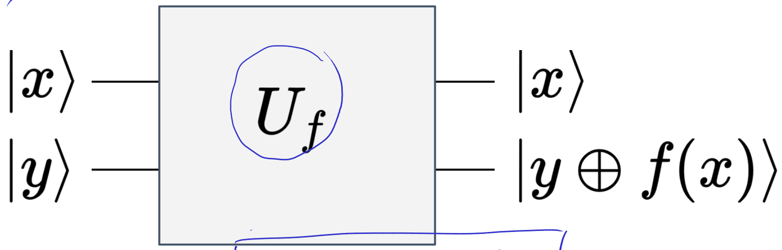
Quantum query complexity

"cost" ≡ # queries made to the oracle.

efficient ≡ lesser # queries. $f_0, f_1, f_x, f_{\bar{x}}$

$$|x, y\rangle \to |x, y \oplus f(x)\rangle$$

$f: \{0,1\} \to \{0,1\}$



"# query" = 1

Deutsch Problem

We are given a balck-box U_f that implements the function f and can evaluate f by querying U_f. We wish to **minimize the number of queries** to U_f to determine some desired property of f.

balanced constant

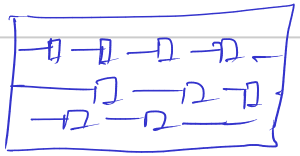
global property

Problem: Determine $\boxed{f(0) \oplus f(1)}$

Our goal is to determine the parity such that the query complexity is reduced. Deutsch algorithm provides an **exponential advantage in terms of query complexity**.

$\boxed{2^1 \to 1} \Rightarrow \boxed{2^n \to 1}$

We assume that the cost of an oracle-query is fixed and are not concerned with the complexity of function evaluation and the details of its implementation.



"unit cost"

⇒ not concerned with gate complexity or circuit depth.

REF: Understanding Quantum Algorithms via Query Complexity, Andris Ambainis

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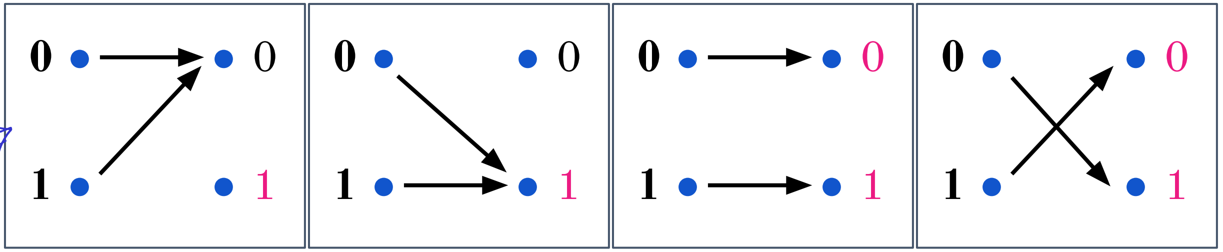
Setting-up the Deutsch Problem

f_0	f_1	f_x	$f_{\bar{x}}$
$f(x) = 0$	$f(x) = 1$	$f(x) = x$	$f(x) = \bar{x}$
$f(0) = 0$ $f(1) = 0$	$f(0) = 1$ $f(1) = 1$	$f(0) = 0$ $f(1) = 1$	$f(0) = 1$ $f(1) = 0$



$f(0) = f(1)$

$f(0) \oplus f(1) = 0$

constant

$f(0) \neq f(1)$

$f(0) \oplus f(1) = 1$

balanced

black-box access
→ function definition is hidden
→ function can be evaluated over inputs from its domain

Deutsch Problem

$\boxed{f(x) : \{0,1\} \rightarrow \{0,1\}}$ **PROMISE**

- f is **guaranteed** to be balanced or constant
- Given a black-box access to f → $f(0), f(1)$
- Determine whether f is **balanced** or **constant**
  - simple, artificial, contrived problem
  - **quantum algorithm provides an exponential advantage over the best classical algorithm**

"Deutsch's algorithm is a perfect illustration of all that is miraculous, subtle, and disappointing about quantum computers. It calculates a solution to a problem faster than any classical ever can. It illustrates the subtle interaction of **superposition**, **phase-kick back**, and **interference**. Finally, is solves a completely pointless problem."

- Mark Oskin, UW

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